Branch Coverage Report

Smart Complaint Management System · Team 5D
Decision (branch) coverage - Part 3

Version	1.0
Date of execution	12 Nov 2026, 15:45
Commit tested	66d1420
Branch	main
Test framework	pytest, pytest-cov (coverage.py), pytest-html
Overall result	114 of 114 automated tests passed

1. Summary

Branch coverage is 86.6%: 265 of 306 branch outcomes were taken. 39 decision points were only partly exercised (one outcome taken, the other not). The combined statement+branch figure reported by coverage.py is 93.1%.

Branch coverage counts each possible outcome of every if, loop and conditional expression. A line can be 100% statement-covered while one of its outcomes is never tested, which is why this report is kept separate from the statement coverage report.

2. Branch coverage by file

File	Branches	Taken	Partial	Branch %	Missed arcs (line -> target)
app.py	2	1	1	50.0	46->47
nlp/categorizer.py	2	2	0	100.0	-
nlp/train_model.py	2	1	1	50.0	30->31
routes/complaint_routes.py	36	27	9	75.0	102->103, 105->120, 106->105, 107->108, 112->116, 141->146, 142->141, 155->166, 156->155
routes/escalation_routes.py	2	2	0	100.0	-
routes/evidence_routes.py	18	15	3	83.3	39->40, 41->42, 50->51
routes/notification_routes.py	2	2	0	100.0	-
routes/resolution_routes.py	8	6	2	75.0	47->48, 49->50
routes/sla_routes.py	6	3	1	50.0	50->51, 65->exit, 65->66
services/analytics.py	12	11	1	91.7	17->18
services/auth.py	26	20	6	76.9	42->45, 51->52, 57->58, 59->60, 105->106, 133->134
services/complaints.py	28	27	1	96.4	35->36
services/escalation.py	22	20	2	90.9	26->27, 84->85
services/evidence.py	20	19	1	95.0	29->30
services/notifications.py	50	45	5	90.0	58->60, 60->61, 116->117, 133->134, 160->exit
services/resolution.py	20	20	0	100.0	-
services/routing.py	4	4	0	100.0	-
services/sla.py	30	28	2	93.3	31->32, 106->107
services/status.py	12	10	2	83.3	36->37, 52->55
services/ticket_id.py	4	2	2	50.0	32->33, 35->36
TOTAL	306	265	39	86.6	

Files with no decision points (for example routes/dashboard_routes.py) are omitted from the table and are fully covered.

3. Analysis of untaken branches

• routes/complaint_routes.py (9 missed arcs): the lookup loops never fall through to 'not found' in tests, and category override is never tried on a complaint whose work has started (status past Assigned), so the 'do not re-route' outcome (line 112 -> 116) is untested.

• services/auth.py (6): the false outcomes of create_user's validation checks listed above, find_user returning nothing, and the disabled-account check on a valid token.

• routes/evidence_routes.py and routes/resolution_routes.py: 404 and 403 outcomes when listing or downloading notes and evidence.

• routes/sla_routes.py: the background monitor loop is never entered, and /sla/alerts is never called by an agent.

• services/ticket_id.py: the daily-limit and duplicate guards are defensive and never true in normal use.

• services/notifications.py: SMTP with TLS turned off and SMTP login with a username are not reached, because tests replace the providers with an in-memory outbox.

• Suggested Sprint 3 additions: not-found and forbidden tests for each complaint endpoint, an override after work has started, create_user validation tests, and a token for a deactivated account. These cover most of the remaining partial branches without changing product code.

4. Limitations

Branch coverage does not show combinations of conditions (MC/DC) or concurrency. For example, every branch of the submission handler is covered, yet concurrent submissions on main still corrupt the data file (defect DEF-01 in the System Validation Report). Branch coverage is a floor, not proof of correctness.

5. Evidence

docs/reports/coverage-html/index.html marks partial branches in yellow; coverage.xml carries the branch-rate per file.